

CUMSHEWA ISLAND, BC, Canada

WMO: 717710

Lat: **53.030N**Lon: **131.601W**Elev: **14**StdP: **101.16**Time Zone: **-8.00 (NAP)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.9	-1.1	-9.7	1.6	-1.3	-7.0	2.1	-0.4	22.6	5.3	20.7	6.3	5.8	330	0.723

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.0	18.8	15.2	17.5	14.6	16.6	14.1	15.9	18.0	15.2	17.1	14.7	16.4	5.0	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.9	10.6	16.8	14.3	10.2	16.1	13.8	9.9	15.6	44.4	18.1	42.6	17.3	41.1	16.5	18.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.1	16.4	14.0	-4.6	21.6	3.6	1.5	-7.2	22.6	-9.3	23.5	-11.2	24.3	-13.8	25.3		
(5)			-6.0	17.0	4.0	0.7	-8.9	17.5	-11.2	17.9	-13.5	18.3	-16.4	18.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	5.4	4.9	5.2	6.9	9.1	11.6	14.1	14.8	13.1	10.0	7.0	5.5
	DBStd	4.13	3.00	2.40	2.37	1.67	2.04	2.09	1.71	1.48	1.50	1.84	2.48	2.60
	HDD10.0	828	144	142	149	94	40	5	0	0	0	22	93	138
	HDD18.3	3412	402	376	408	343	286	203	133	109	158	258	341	397
	CDD10.0	459	0	0	0	1	13	53	126	149	93	22	2	0
	CDD18.3	2	0	0	0	0	0	1	1	1	0	0	0	0
	CDH23.3	1	0	0	0	0	0	1	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.6	7.0	6.4	6.3	5.9	5.1	4.6	4.0	4.1	4.8	5.7	6.5	6.9
Precipitation	PrecAvg	2262	276	204	196	146	105	73	61	91	147	309	301	317
	PrecMax	2715	469	394	299	222	238	154	135	228	313	494	540	672
	PrecMin	1942	142	75	117	55	47	31	24	34	47	152	159	197
	PrecStd	219	80	79	55	37	47	33	30	49	57	85	83	103
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.3	10.3	10.7	12.8	15.6	19.1	20.8	20.6	17.9	15.6	12.7	11.0
		MCWB	9.1	8.6	8.4	8.2	11.0	13.8	15.5	15.8	14.1	12.9	10.3	9.0
	2%	DB	10.1	9.3	9.5	10.8	13.8	16.4	18.5	19.1	16.6	13.9	11.3	10.0
		MCWB	8.7	8.0	7.7	7.7	10.6	13.2	15.2	15.3	13.7	11.9	9.7	8.5
	5%	DB	9.4	8.7	8.8	9.8	12.6	15.0	17.2	17.8	15.7	12.9	10.7	9.4
		MCWB	8.2	7.5	7.2	7.6	10.0	12.8	14.4	14.9	13.3	11.1	9.3	8.1
	10%	DB	8.8	8.1	8.2	9.1	11.6	14.0	16.2	16.9	15.1	12.2	10.1	8.8
		MCWB	7.8	6.9	6.7	7.3	9.5	12.2	13.8	14.5	12.9	10.6	8.8	7.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	9.5	9.2	9.5	12.3	14.9	16.7	16.9	15.5	13.4	10.9	9.5
		MCDB	10.4	10.1	10.2	10.2	14.4	16.9	19.0	19.5	17.3	14.6	11.4	10.3
	2%	WB	9.0	8.5	8.2	8.7	11.2	14.0	15.7	16.1	14.6	12.5	10.0	8.8
		MCDB	9.6	9.1	9.2	9.7	13.0	15.9	17.9	18.2	15.9	13.5	10.7	9.5
	5%	WB	8.4	7.8	7.7	8.1	10.6	13.2	15.0	15.6	14.0	11.7	9.5	8.3
		MCDB	9.1	8.5	8.7	9.2	12.0	14.9	16.9	17.5	15.2	12.6	10.3	9.1
	10%	WB	7.8	7.2	7.0	7.6	10.0	12.4	14.4	15.0	13.5	11.1	8.9	7.8
		MCDB	8.6	8.0	8.0	8.8	11.2	13.8	16.0	16.8	14.7	12.0	9.9	8.7
Mean Daily Temperature Range	5% DB	MDBR	2.9	3.0	3.2	3.1	3.0	2.9	2.9	3.0	3.0	3.1	3.2	3.1
		MCDBR	3.1	3.2	3.5	3.7	4.3	4.6	4.5	4.4	3.9	3.8	3.3	3.3
		MCWBR	2.7	2.9	2.9	2.7	2.8	2.8	2.6	2.7	2.7	3.0	3.1	2.9
	5% WB	MCDBR	3.0	3.1	3.4	3.1	3.7	4.0	4.2	4.0	3.3	3.6	3.3	3.1
		MCWBR	2.7	2.9	2.9	2.7	2.5	2.6	2.5	2.7	2.6	3.1	3.1	2.9
Clear-Sky Solar Irradiance	taub		0.319	0.325	0.350	0.388	0.386	0.373	0.374	0.374	0.368	0.351	0.328	0.309
	taud		2.372	2.464	2.407	2.286	2.357	2.447	2.487	2.500	2.476	2.417	2.325	2.323
	Ebn at Noon		660	789	837	842	864	879	871	852	811	738	630	590
	Edh at Noon		59	72	92	117	115	106	101	94	86	74	60	51
All-Sky Solar Radiation	RadAvg		0.59	1.43	2.54	3.99	5.05	5.28	5.04	4.27	2.94	1.65	0.78	0.49
	RadStd		0.07	0.23	0.29	0.20	0.40	0.39	0.49	0.35	0.29	0.19	0.08	0.07